21. Java ArrayList Methods — Quick Notes

add()

Add at End

list.add("Geeks");
Adds element at: END
Example: [Geeks, IDE]
After: list.add("Courses");
[Geeks, IDE, Courses]

Add at Index

list.add(1, "Practice");

Before: [Geeks, IDE, Courses]
After: [Geeks, Practice, IDE, Courses]

Important

Elements are shifted right.
Time Complexity: O(n)

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size()

Purpose: Returns number of elements.

list.size();
Example: [10,20,30]

Output: 3
Time Complexity: O(1)

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contains()

Purpose: Checks whether element exists.
list.contains("IDE");

Output: true

If not found: false

Time Complexity: O(n)

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remove()

Remove by Index

list.remove(1);
Before: [Geeks, IDE, Courses]
After: [Geeks, Courses]

Returns: String removed = list.remove(1);
Output: IDE

Remove by Object
list.remove("Courses");
Returns: true

if removed.
Returns: false

if not present.

Difference

Method	Returns
remove(index)	Removed Element
remove(object)	boolean

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get()
Purpose: Get element at index.
list.get(1);
Example: [10,20,30]

Output: 20
Time Complexity: O(1)

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set()
Purpose: Replace element at index.
list.set(1, 40);

Before: [10,20,30]
After: [10,40,30]

Time Complexity: O(1)

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indexOf()
Purpose: Returns first occurrence index.

list.indexOf(10);
Example: [10,20,10,30]

Output: 0
Returns: -1 if not found.

Time Complexity:O(n)

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lastIndexOf()
Purpose: Returns last occurrence index.

list.lastIndexOf(10);

Example: [10,20,10,30]
Output: 2
Returns:-1 if not found.

Time Complexity: O(n)

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clear()
Purpose: Removes all elements.
list.clear();
Before: [10,20,30]
After: []

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isEmpty()
Purpose: Checks whether list is empty.

list.isEmpty();

Output: true or false
Time Complexity: O(1)

Important Index Rules
For: Size = 3
Valid indexes: 0,1,2

For insertion: list.add(index, item);
Valid: 0 to size

Example: list.add(3, 40); Adds at end.
Invalid: list.add(4, 40);

Throws: IndexOutOfBoundsException

Time Complexity Summary

Operation	Complexity
add(e)	Amortized O(1)
add(index,e)	O(n)
size()	O(1)
isEmpty()	O(1)
get()	O(1)
set()	O(1)
contains()	O(n)
indexOf()	O(n)
lastIndexOf()	O(n)
remove(index)	O(n)
remove(object)	O(n)
clear()	O(n)

Most Important Interview Points

Method	Key Point
add(e)	Adds at end
add(index,e)	Shifts elements right
get(i)	Random Access O(1)
set(i,e)	Replace element
contains()	Linear Search
indexOf()	First occurrence
lastIndexOf()	Last occurrence
remove(index)	Returns removed element
remove(object)	Returns boolean
size()	Actual element count
isEmpty()	Checks size == 0